

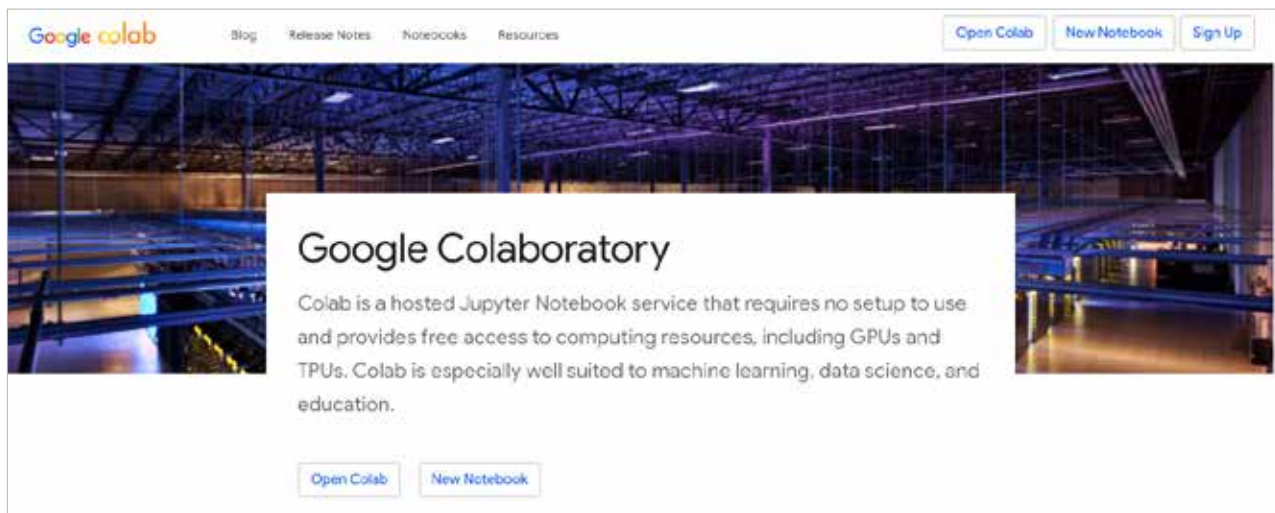


Figure 1: Google Colab

The first step is to import the required libraries. We will use the Pandas and Sklearn libraries to build the machine learning model. You can use the following code for this:

```
import sklearn
import pandas as pd
import numpy as np
```

If you don't have the libraries already installed, use the following command:

```
pip install "library name"
```

I have taken a sample dataset that has a few of these categories, like name, credit card number, phone number, date of birth, etc. You can create this on your own using different formats of the category. Then load the dataset and view it as shown below and in Figure 2.

```
df = pd.read_csv("//content/drive/MyDrive/data.csv")
df = df.sample(frac = 1)
df.head()
```

	key	value	label	final
8310	33517752	accountnumber	pii	account number
7442	2123 2681 6525 0328	cc	pii	credit card
4703	10/03/25	birthday	pii	birthday
8482	36824119	bankaccount	pii	account number
9904	7158319164	bankaccount	pii	account number

Figure 2: Dataset

The first step after collecting and building the dataset

in the data science life cycle is to extract or identify the features. The first feature we are using here is the length of the string. We can do this using the following code, as shown in Figure 3.

```
df["noofcharkey"]=df["key"].str.len()
df.head()
```

	key	value	label	final	noofcharkey
8310	33517752	accountnumber	pii	account number	8
7442	2123 2681 6525 0328	cc	pii	credit card	19
4703	10/03/25	birthday	pii	birthday	8
8482	36824119	bankaccount	pii	account number	8
9904	7158319164	bankaccount	pii	account number	10

Figure 3: Length of the string

Similarly, we can extract a few more features – for example, if the string has an '@' in it, contains spaces, has a '/', is a number, is a decimal, and so on. The following code can be used to do this, and our dataset then looks as shown in Figure 4.

```
abc=pd.DataFrame(df['key'].str.contains("@", case=False))
abc.head()
list1=abc["key"]
df["key_has_@"]=list1
abc=pd.DataFrame(df['value'].str.contains(" ", case=False))
abc.head()
list1=abc["value"]
df["space_in_value"]=list1
d={'noofcharkey': 'noofchar_key'}
```